CARRIER — instrument side (generated by gen_carrier_sch.py; do not edit by hand). Everything here is powered by the Pico's own USB; nothing is fed from the DUT.

INSTR_3V3 comes only from Pico pin 36 (3V3_OUT). Pico VBUS (40) and VSYS (39) are left unconnected. Pico, MikroE and microSD plug into Harwin sockets J1–J5.

J2 pin k = Pico pin 20+k; J4 pin k = MikroE pin 8+k. Physical socket order is fixed at footprint stage (Pico pin 21 is opposite pin 20).

Sense: J7 -> C1 (100n across the Kelvin pair, close to the module) -> J6 wire pads -> MikroE IN1+ / IN1− / IN2 VBUS screw terminals. J7 pin 4 is the shared ground reference.

J1
PICO_UART_TX 1
PICO_UART_RX 2
INSTR_GND 3
PICO_MARK_RX 4
UART_OE_N 5
GP4 (unused) 6
GP5 (unused) 7
INSTR_GND 8
SD_CARD_PRESENT 9
SD_CS_N 10
SD_MISO 11
PM_ALERT_N 12
INSTR_GND 13
SD_SCLK 14
SD_MOSI 15
PM_SDA 16
PM_SCL 17
INSTR_GND 18
GP14 (unused) 19
GP15 (unused) 20
Pico pins 1–20
J_PICO_LEFT

J3
AN (unused) 1
RST (unused) 2
CS (unused) 3
SCK (unused) 4
MISO (unused) 5
MOSI (unused) 6
INSTR_3V3 7
INSTR_GND 8
MikroE pins 1–8
J_PM_LEFT

J6
SHUNT_KELVIN_PLUS 1
SHUNT_KELVIN_MINUS 2
DUT_VBUS_SENSE 3
Wire pads to MikroE IN1+/IN1−/IN2 VBUS
J_PM_ANALOG

J7
SHUNT_KELVIN_PLUS 1
SHUNT_KELVIN_MINUS 2
DUT_VBUS_SENSE 3
INSTR_GND 4
Sense from interposer
J_SENSE

C1
SHUNT_KELVIN_PLUS — SHUNT_KELVIN_MINUS
100n X7R 50V
C_SENSE_DIFF

J4
INSTR_GND 1
5V (unused) 2
PM_SDA 3
PM_SCL 4
TX (unused) 5
RX (unused) 6
PM_ALERT_N 7
PWM (unused) 8
MikroE pins 9–16
J_PM_RIGHT

Indicators: Pico GP28, GP27, GP26, GP22, GP21 (LED1..LED5) -> 1.8k -> red LED -> ground.

R6 LED1_DRIVE — LED1_A
1k8 1%
R_LED1

D1 LED1_A — INSTR_GND
TLLR4401 red
LED1

R7 LED2_DRIVE — LED2_A
1k8 1%
R_LED2

D2 LED2_A — INSTR_GND
TLLR4401 red
LED2

R8 LED3_DRIVE — LED3_A
1k8 1%
R_LED3

D3 LED3_A — INSTR_GND
TLLR4401 red
LED3

R9 LED4_DRIVE — LED4_A
1k8 1%
R_LED4

D4 LED4_A — INSTR_GND
TLLR4401 red
LED4

R10 LED5_DRIVE — LED5_A
1k8 1%
R_LED5

D5 LED5_A — INSTR_GND
TLLR4401 red
LED5

J2
GP16 (unused) 1
GP17 (unused) 2
INSTR_GND 3
GP18 (unused) 4
GP19 (unused) 5
GP20 (unused) 6
LED5_DRIVE 7
INSTR_GND 8
LED4_DRIVE 9
RUN (unused) 10
LED3_DRIVE 11
LED2_DRIVE 12
AGND (unused) 13
LED1_DRIVE 14
ADC_VREF (unused) 15
INSTR_3V3 16
3V3_EN (unused) 17
INSTR_GND 18
VSYS (unused) 19
VBUS (unused) 20
Pico pins 21–40
J_PICO_RIGHT

J5
INSTR_3V3 1
INSTR_GND 2
SD_SCLK 3
SD_MISO 4
SD_MOSI 5
SD_CS_N 6
SD_DAT1 (unused) 7
SD_DAT2 (unused) 8
SD_CARD_PRESENT 9
Adafruit 4682 microSD
J_SD

SN74LV125AN
U_UART
U1A: PI_TX_CABLE 2 — PICO_UART_RX 3; UART_OE_N 1
U1B: PI_MARK_CABLE 5 — PICO_MARK_RX 6; UART_OE_N 4
U1C: PICO_UART_TX 9 — PICO_TX_DRV 8; UART_OE_N 10
U1D: INSTR_GND 12 — 11; INSTR_3V3 13
U1E: VCC 14 = INSTR_3V3; GND 7 = INSTR_GND

C2
INSTR_3V3 — INSTR_GND
100n X7R 50V
C_UART

R2 PI_TX_CABLE — INSTR_GND
100k 1%
R_PI_TX_PD

R3 PI_MARK_CABLE — INSTR_GND
100k 1%
R_PI_MARK_PD

R1 PICO_TX_DRV — PI_RX_CABLE
470 1%
R_PICO_TX

R4 UART_OE_N — INSTR_3V3
10k 1%
R_UART_OE_PU

R5 PICO_UART_TX — INSTR_3V3
10k 1%
R_PICO_TX_PU

J8
INSTR_GND 1
PI_TX_CABLE 2
PI_RX_CABLE 3
PI_MARK_CABLE 4
5
UART/marker from interposer
J_UART

UART (docs/uart-interface-qualification.md): U1 is USB-powered. Keep UART_OE_N high (R4) until the Pico has set up its pins. Pico TX is normal polarity (R5 idle-high).

R2/R3 load the Pi outputs (approved UART-bias exception). R1 is the 470R source resistor for Pico->Pi. U1 channel 4 unused: input low, enable high.

Prototype — not released
picopower-proto
Sheet: /
File: carrier.kicad_sch
Title: Carrier
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.9 | Id: 1/1